REFLEX Fixed Methods and Seed Results

All fixed methods and seeds are retained below. Raw threshold counts differ from final policy dispositions when review is possible. No method or seed is selected from these results. TP denotes a confirmed defect decision; FC denotes automatic false clear; FP denotes a final false-defect alert. Unresolved reviews are excluded from confirmed detections.

The independent image oracle verifies 88 method/group rows and 9,460 predictions over 860 unique population cases. Repeated arms are paired evaluations, not additional independent observations.

candle replay

Recorded method	AUROC	AP	Raw TP FP FN TN	Final TP FP FC TN	Reviews
classical_cv5	0.7575	0.7313	16 / 4 / 84 / 96	16 / 4 / 84 / 96	0
dense128_max	0.9489	0.9535	69 / 3 / 31 / 97	69 / 3 / 21 / 94	13
dense128_pose	0.9441	0.9486	71 / 4 / 29 / 96	71 / 4 / 29 / 96	0
dense128_region	0.9566	0.9546	69 / 3 / 31 / 97	78 / 6 / 18 / 94	4
dense128_seed20261025	0.9476	0.9509	75 / 6 / 25 / 94	75 / 6 / 25 / 94	0
dense128_seed20261026	0.9298	0.9317	64 / 3 / 36 / 97	64 / 3 / 26 / 94	13
dense128_top4	0.9519	0.9537	72 / 3 / 28 / 97	72 / 3 / 19 / 93	13
dense512_max	0.9373	0.9374	73 / 6 / 27 / 94	73 / 6 / 19 / 92	10
dense512_seed20261025	0.9273	0.9332	70 / 4 / 30 / 96	70 / 4 / 23 / 92	11
dense512_seed20261026	0.9375	0.9409	70 / 5 / 30 / 95	70 / 5 / 30 / 95	0
v1_recomputed_metrics	0.9005	0.8967	42 / 3 / 58 / 97	42 / 3 / 58 / 97	0

cashew replay

Recorded method	AUROC	AP	Raw TP FP FN TN	Final TP FP FC TN	Reviews
classical_cv5	0.5964	0.7124	5 / 3 / 95 / 47	5 / 3 / 95 / 47	0
dense128_max	0.9512	0.9778	73 / 0 / 27 / 50	73 / 0 / 27 / 50	0
dense128_pose	0.9650	0.9841	75 / 0 / 25 / 50	75 / 0 / 10 / 43	22
dense128_region	0.9600	0.9818	77 / 0 / 23 / 50	77 / 0 / 23 / 50	0
dense128_seed20261025	0.9606	0.9817	77 / 1 / 23 / 49	77 / 1 / 23 / 49	0
dense128_seed20261026	0.9564	0.9798	72 / 0 / 28 / 50	72 / 0 / 28 / 50	0
dense128_top4	0.9582	0.9810	74 / 0 / 26 / 50	74 / 0 / 26 / 50	0
dense512_max	0.9636	0.9824	74 / 1 / 26 / 49	74 / 1 / 26 / 49	0
dense512_seed20261025	0.9488	0.9772	73 / 0 / 27 / 50	73 / 0 / 27 / 50	0
dense512_seed20261026	0.9478	0.9767	76 / 0 / 24 / 50	76 / 0 / 24 / 50	0
v1_recomputed_metrics	0.9476	0.9744	70 / 2 / 30 / 48	70 / 2 / 30 / 48	0

capsules development

Recorded method	AUROC	AP	Raw TP FP FN TN	Final TP FP FC TN	Reviews
classical_cv5	0.6087	0.7197	5 / 2 / 45 / 28	5 / 2 / 45 / 28	0
dense128_max	0.6800	0.8029	14 / 2 / 36 / 28	14 / 2 / 36 / 28	0
dense128_pose	0.6593	0.7837	10 / 1 / 40 / 29	10 / 1 / 36 / 28	5
dense128_region	0.6680	0.7947	14 / 2 / 36 / 28	14 / 2 / 36 / 28	0
dense128_seed20261025	0.6880	0.8087	16 / 1 / 34 / 29	16 / 1 / 30 / 27	6
dense128_seed20261026	0.6727	0.8081	16 / 3 / 34 / 27	16 / 3 / 29 / 27	5
dense128_top4	0.6720	0.7942	13 / 2 / 37 / 28	13 / 2 / 37 / 28	0
dense512_max	0.6593	0.7943	19 / 4 / 31 / 26	19 / 4 / 31 / 26	0
dense512_seed20261025	0.6747	0.8060	11 / 1 / 39 / 29	11 / 1 / 39 / 29	0
dense512_seed20261026	0.7013	0.8182	13 / 2 / 37 / 28	13 / 2 / 37 / 28	0
v1_fresh	0.7027	0.8137	16 / 1 / 34 / 29	16 / 1 / 34 / 29	0

capsules final

Recorded method	AUROC	AP	Raw TP FP FN TN	Final TP FP FC TN	Reviews
classical_cv5	0.5653	0.6971	9 / 4 / 41 / 26	9 / 4 / 41 / 26	0
dense128_max	0.6813	0.8345	17 / 1 / 33 / 29	17 / 1 / 33 / 29	0
dense128_pose	0.7540	0.8648	13 / 0 / 37 / 30	13 / 0 / 31 / 30	6
dense128_region	0.6713	0.8278	16 / 1 / 34 / 29	16 / 1 / 34 / 29	0
dense128_seed20261025	0.6927	0.8459	18 / 0 / 32 / 30	18 / 0 / 27 / 30	5
dense128_seed20261026	0.7067	0.8351	16 / 0 / 34 / 30	16 / 0 / 30 / 28	6
dense128_top4	0.6747	0.8300	17 / 1 / 33 / 29	17 / 1 / 33 / 29	0
dense512_max	0.7253	0.8414	19 / 1 / 31 / 29	19 / 1 / 31 / 29	0
dense512_seed20261025	0.7207	0.8513	14 / 0 / 36 / 30	14 / 0 / 36 / 30	0
dense512_seed20261026	0.6567	0.8191	14 / 0 / 36 / 30	14 / 0 / 36 / 30	0
v1_fresh	0.7233	0.8421	16 / 0 / 34 / 30	16 / 0 / 34 / 30	0

chewinggum development

Recorded method	AUROC	AP	Raw TP FP FN TN	Final TP FP FC TN	Reviews
classical_cv5	0.5520	0.6786	1 / 1 / 49 / 24	1 / 1 / 49 / 24	0

Recorded method	AUROC	AP	Raw TP FP FN TN	Final TP FP FC TN	Reviews
dense128_max	0.9920	0.9965	49 / 3 / 1 / 22	49 / 3 / 1 / 19	3
dense128_pose	0.9904	0.9958	48 / 0 / 2 / 25	48 / 0 / 2 / 23	2
dense128_region	0.9912	0.9964	49 / 1 / 1 / 24	49 / 1 / 1 / 21	3
dense128_seed20261025	0.9864	0.9940	47 / 0 / 3 / 25	47 / 0 / 3 / 25	0
dense128_seed20261026	0.9984	0.9992	48 / 1 / 2 / 24	48 / 1 / 1 / 24	1
dense128_top4	0.9920	0.9967	49 / 1 / 1 / 24	49 / 1 / 1 / 24	0
dense512_max	0.9808	0.9928	48 / 2 / 2 / 23	48 / 2 / 2 / 23	0
dense512_seed20261025	0.9880	0.9948	48 / 3 / 2 / 22	48 / 3 / 2 / 22	0
dense512_seed20261026	0.9936	0.9971	48 / 1 / 2 / 24	48 / 1 / 2 / 24	0
v1_fresh	0.9680	0.9885	48 / 0 / 2 / 25	48 / 0 / 2 / 25	0

chewinggum final

Recorded method	AUROC	AP	Raw TP FP FN TN	Final TP FP FC TN	Reviews
classical_cv5	0.5736	0.7069	3 / 0 / 47 / 25	3 / 0 / 47 / 25	0
dense128_max	0.9744	0.9905	48 / 4 / 2 / 21	48 / 4 / 2 / 20	1
dense128_pose	0.9944	0.9972	47 / 2 / 3 / 23	47 / 2 / 1 / 23	2
dense128_region	0.9736	0.9894	48 / 4 / 2 / 21	48 / 4 / 2 / 20	1
dense128_seed20261025	0.9848	0.9938	48 / 2 / 2 / 23	48 / 2 / 2 / 23	0
dense128_seed20261026	0.9840	0.9931	47 / 2 / 3 / 23	47 / 2 / 1 / 22	3
dense128_top4	0.9712	0.9890	48 / 3 / 2 / 22	48 / 3 / 2 / 22	0
dense512_max	0.9768	0.9909	47 / 2 / 3 / 23	47 / 2 / 3 / 23	0
dense512_seed20261025	0.9840	0.9933	48 / 3 / 2 / 22	48 / 3 / 2 / 22	0
dense512_seed20261026	0.9880	0.9954	49 / 0 / 1 / 25	49 / 0 / 1 / 25	0
v1_fresh	0.9840	0.9929	47 / 3 / 3 / 22	47 / 3 / 3 / 22	0

pcb1 development

Recorded method	AUROC	AP	Raw TP FP FN TN	Final TP FP FC TN	Reviews
classical_cv5	0.7916	0.7713	12 / 2 / 38 / 48	12 / 2 / 38 / 48	0
dense128_max	0.9176	0.9171	34 / 3 / 16 / 47	34 / 3 / 16 / 47	0
dense128_pose	0.9172	0.9119	32 / 4 / 18 / 46	32 / 4 / 13 / 44	7
dense128_region	0.9112	0.9101	33 / 6 / 17 / 44	38 / 6 / 7 / 42	7

Recorded method	AUROC	AP	Raw TP FP FN TN	Final TP FP FC TN	Reviews
dense128_seed20261025	0.8740	0.8793	37 / 6 / 13 / 44	37 / 6 / 12 / 40	5
dense128_seed20261026	0.8856	0.8948	35 / 4 / 15 / 46	35 / 4 / 12 / 45	4
dense128_top4	0.9248	0.9226	31 / 5 / 19 / 45	31 / 5 / 19 / 45	0
dense512_max	0.8912	0.8976	30 / 4 / 20 / 46	30 / 4 / 20 / 46	0
dense512_seed20261025	0.9080	0.9169	33 / 3 / 17 / 47	33 / 3 / 17 / 47	0
dense512_seed20261026	0.9200	0.9216	33 / 3 / 17 / 47	33 / 3 / 17 / 47	0
v1_fresh	0.8932	0.8901	30 / 3 / 20 / 47	30 / 3 / 20 / 47	0

pcb1 final

Recorded method	AUROC	AP	Raw TP FP FN TN	Final TP FP FC TN	Reviews
classical_cv5	0.7848	0.7412	10 / 4 / 40 / 46	10 / 4 / 40 / 46	0
dense128_max	0.8688	0.8629	33 / 5 / 17 / 45	33 / 5 / 17 / 45	0
dense128_pose	0.8916	0.8841	34 / 5 / 16 / 45	34 / 5 / 8 / 42	11
dense128_region	0.8780	0.8733	38 / 8 / 12 / 42	41 / 9 / 6 / 34	10
dense128_seed20261025	0.8792	0.8872	36 / 7 / 14 / 43	36 / 7 / 9 / 40	8
dense128_seed20261026	0.8512	0.8607	34 / 4 / 16 / 46	34 / 4 / 11 / 40	11
dense128_top4	0.8856	0.8794	36 / 5 / 14 / 45	36 / 5 / 14 / 45	0
dense512_max	0.8596	0.8544	30 / 6 / 20 / 44	30 / 6 / 20 / 44	0
dense512_seed20261025	0.8556	0.8721	33 / 5 / 17 / 45	33 / 5 / 17 / 45	0
dense512_seed20261026	0.8608	0.8708	32 / 3 / 18 / 47	32 / 3 / 18 / 47	0
v1_fresh	0.8488	0.8539	29 / 6 / 21 / 44	29 / 6 / 21 / 44	0

Interpretation

Seed 20261024 defines the original memory construction. Seeds 20261025 and 20261026 are the fixed repeats, not post-test choices. The classical arm is a practical baseline with different internal resolution. v1_fresh preserves the older algorithm fitted on the new category membership; v1_recomputed_metrics retains earlier replay predictions while correcting the historical AU-PRO definition. The complete runtime, pixel, native-coordinate, crop-control, robustness and reference exports retain their own scopes and verification receipts.

Policy contribution accounting

A post-evaluation descriptive audit of unchanged outcomes finds identical false-clear case-ID sets for review-only, chosen-crop and random-crop paths in every group. The review band removes those automatic passes; crop choice does not further reduce them. Across 46 executed active cases (29 defective and 17 normal), chosen detail resolves 17 additional true alerts and 4 false alerts relative to review without ROI. Random detail resolves 12 true alerts and 2 false alerts. Review-only leaves 46 unresolved, chosen 25, and random 32; all three retain 127 automatic false clears in the execution audit. These cross-population totals audit controller behavior, not pooled generalization or a superiority test.